Local Projections in Macroeconomics

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Short-run zero restrictions

- Consider the classical identification of the MP shocks (see, e.g., Christiano et al. (1999)).
- Assumptions: on impact a set of variables do not move (call them slow moving), whereas others do (call them fast moving).
- This can be translated into LPs.

LP regression

Consider the following LP, where i_t denotes the interest rate:

$$y_{t+h} = \alpha_h + \beta_h i_t + \kappa_h y_t^S + \gamma_h x_t + u_{t+h}$$

where x_t includes lags of y_t, i_t, y_t^S, and potentially other controls.
Caveat: control contemporaneously only for the slow moving variables y_t^S.

Intuition – OLS recap I

Consider the simple univariate linear regression model:

$$z_t^1 = b_0 + b_1 z_t^2 + \epsilon_t$$

What does b_1 measures? Formally,

$$b_1 = \frac{\text{cov}[z_t^1, z_t^2]}{\text{var}[z_t^2]}$$

That is, how does a unit shift in z_t^2 affects z_t^1.

Intuition – OLS recap II

Consider now the multivariate extension:

$$z_t^1 = b_0 + b_1 z_t^2 + b_2 z_t^3 + \epsilon_t$$

What does b_1 measures? Formally,

$$b_1 = \frac{\text{cov}[(z_t^1|z_t^3),(z_t^2|z_t^3)]}{\text{var}[z_t^2|z_t^3]} = \frac{\text{cov}[(z_t^1 \perp z_t^3),(z_t^2 \perp z_t^3)]}{\text{var}[z_t^2 \perp z_t^3]}$$

That is, how does a unit shift in z_t^2 affects z_t^1 holding fixed z_t^3.

Intuition – Back to LPs

$$y_{t+h} = \alpha_h + \beta_h i_t + \kappa_h y_t^S + \gamma_h x_t + u_{t+h}$$

$$\Longleftrightarrow$$

$$\left(y_{t+h} \perp y_t^S, x_t\right) = \alpha_h + \beta_h \left(i_t \perp y_t^S, x_t\right) + u_{t+h}$$

Then, what does β_h identify? How a unit increase in the interest rate, **holding fixed all slow moving variables**, affects y_{t+h}.
$\Rightarrow \mathcal{R}_{\text{MP}\rightarrow y}(h)$ h-periods ahead response of y to a monetary policy shock (if you believe to the assumptions of the classical identification strategies).

Long-run zero restrictions

Setting:

- y_t: log real GDP
- u_t: unemployment rate
- $w_t = (y_t, u_t)'$

Blanchard and Quah (1989) identification strategy:

- The long-run effect of a demand shock on the level of output is zero.

We translate it into LPs.

Two-step procedure – supply shock

1. Calculate the linear combination that most affects output in the long run:

$$y_{t+H} - y_{t-1} = \mu_H + g_H w_t + \sum_{j=1}^{p} \phi_j w_{t-j} + \eta_{t+H} \qquad H - \text{large}$$

2. Calculate the LP using $g_H \hat{w}_t$ as a signal of the supply shock:

$$w_{i,t+h} - w_{i,t-1} = \alpha_h + \beta_{i,h}(\hat{g}_H w_t) + \sum_{j=1}^{p} \gamma_j w_{t-j} + \eta_{t+h}$$

for $i = y, u$.

$\Rightarrow$ $\beta_{i,h}$ h-periods ahead response of $w_{i,t} = \{y_t, u_t\}$ to a supply shock.

Two-step procedure – demand shock

1. As before.
2. Impose that demand shocks are orthogonal to $\hat{g}_H w_t$:

$$w_{i,t+h} - w_{i,t-1} = \alpha_h + b_{i,h} u_t + \gamma_0 (\hat{g}_H w_t) + \sum_{j=1}^{p} \gamma_j w_{t-j} + \eta_{t+h}$$

for $i = y, u$.

$\Rightarrow$ $b_{i,h}$ h-periods ahead response of $w_{i,t} = \{y_t, u_t\}$ to a demand shock.